

Vincent (Dam Duc Huy) Ngo

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ACADEMIC & CO-OP STATUS

Computer Engineering, BSc Co-op, University of Alberta

Class of 2029

Cumulative Grade Point Average

3.8/4.0

Completed Academic Terms

3 of 8

Availability Starting May 2026

4 months

TECHNICAL SKILLS

Core Programming & Foundations

- **Languages:** C, C++, Embedded C, Rust, Assembly for ARM, Python, Bash.
- **System Programming:** Multithreaded systems, CLI tools, POSIX/Linux environments.
- **Technical Foundations:** Data Structures & Algorithms, Signal Processing, Real-Time Systems.

Embedded Systems & Hardware

- **Microcontrollers:** STM32 (ARM Cortex-M), ESP32, ATmega328P.
- **Embedded Development:** Bare-metal programming, peripheral drivers, hardware–firmware integration.
- **Hardware Peripherals:** GPIO, ADC, DMA, Timers, Interrupts (ISR), Flash, JTAG, SWD.
- **Real-Time OS:** FreeRTOS (task scheduling, queues, synchronization).
- **Single Board Computers:** Raspberry Pi 4, Zybo Z7-20 (ARM Cortex-A9), Xiphos Q7S (AlbertaSat).

Communication Protocols

- **Embedded Interfaces:** UART, SPI, I²C, CAN, I2S.
- **Debugging & Integration:** Hardware bring-up, firmware debugging, protocol testing

Embedded Linux & Systems

- **Embedded Linux:** Buildroot, Yocto Project, PetaLinux.
- **System Development:** Ground station tools, system automation, firmware utilities.

Digital Design & FPGA

- **FPGA Development:** VHDL, Verilog.
- **Tools:** Vivado.

Tools & Development Environment

- **Development Tools:** ESP-IDF, CMSIS-DSP, MATLAB, Git, GDB (debugging).
- **Hardware Tools:** Oscilloscope, logic analyzer, soldering.
- **PCB Design:** Altium Designer, KiCAD.

TECHNICAL EXPERIENCE & PROJECTS

Flight Software & Embedded Systems Team Member

Jul 2025 – Present

AlbertaSat

- Collaborated on the development of the Ex-Altia 3 CubeSat mission, focusing on the On-Board Computer (OBC) running **embedded Linux (Yocto/PetaLinux), RTOS, and BIOS-level configuration**.
- Performed multiple **hardware - firmware integration tests** in AlbertaSat's clean room, validating and debugging hardware integration over **UART, SPI, I²C, and CAN protocols**.
- Developed **Rust-based flight software handlers**, ground station CLI tools, and automation scripts to streamline mission operations and telemetry handling.

Smartwatch Health Monitor - Hackathon Project

Feb 2026

Embedded Systems Lead | ESP32 | ESP-IDF | FreeRTOS

- **Led a 5-member engineering team** to architect and prototype a wearable health monitoring system within a 48-hour hackathon, coordinating firmware, hardware, and mechanical integration.
- Developed embedded firmware for ESP32 using ESP-IDF and FreeRTOS, leveraging the dual-core architecture to separate real-time sensor acquisition from UI and system management tasks.
- Integrated multiple biometric and environmental sensors over I²C, including MAX30102 (heart rate/SpO₂), MPU6500 (motion & fall detection), ENS160 (air quality), and AHT21 (temperature/humidity).
- Collaborated in custom PCB design and hardware bring-up, supporting schematic review, prototype assembly, and firmware debugging of peripheral interfaces.
- Delivered a fully integrated wearable prototype with real-time health monitoring and on-device visualization inside a 3D-printed smartwatch enclosure within the hackathon deadline.

LED Audio Visualizer

May – Jul 2025

Personal Project

- Designed and programmed an **STM32F103C8 - based embedded system** to visualize live audio signals on a 60 LED strip.
- Implemented **CMSIS-DSP Fast Fourier Transform (FFT)** to analyze 0–20 kHz audio spectrum in real time with less than 20 ms latency.
- Developed ADC peripherals for audio sampling and optimized memory footprint to fit within **20 KB SRAM constraints**.
- Integrated low-level GPIO control, timer interrupts, and DMA to ensure smooth LED updates without blocking CPU tasks.

OTHER WORK EXPERIENCE

Co-Manager

Feb 2022 – Jan 2024

Spunky's Convenience Store, Regina, SK

- Co-managed daily operations of a family-owned store and U-Haul dealership, overseeing cashiering, inventory, and equipment coordination.
- Coordinated with U-Haul systems and corporate processes to manage reservations and ensure dealership standards.

Community Project Lead

September 2025 – Present

Edmonton Viets Association, Edmonton, AB

- Organized a major fundraising initiative for flood relief in Vietnam, managing digital payment platforms and coordinating community donations.
- Collaborated on large-scale community events, including the Heritage Festival, serving as a key liaison for association leadership.

SCHOLARSHIPS & AWARDS

The Robert Stollery Scholarship in Engineering	2024 - 2025
The Albert Julian Hrapko Memorial Scholarship in Computer Engineering	2025
Third Place - Natignite 2025 Hackathon	2025

LEADERSHIP & COMMUNITY INVOLVEMENT

Outreach Volunteer, AlbertaSat	2025
DiscoverE Summer Camp Outreach, AlbertaSat	2025
Secretary General, Edmonton Vietnamese Association	2025

ADDITIONAL INFORMATION

Class 5 Driver's License – Clean abstract

Willing to relocate

Languages – English (Fluent), Vietnamese (Fluent), French (Basic)